Source Data, Code, and Reproducibility Guide

End-to-end autonomous scientific discovery on a real optical platform

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Purpose

This guide accompanies the Zenodo source-data archive for the manuscript *End-to-end autonomous scientific discovery on a real optical platform*. We provide it to help reviewers and readers locate the data, offline analysis scripts, generated figures, provenance records, and report-level material supporting the analyses reported in the paper.

The archive is arranged in five report-based sections, with additional files for selected main-text data-panel replots and a short qualitative video of the optical platform. The formal main manuscript and the formal Supplementary Information are not duplicated here; they are submitted separately through the journal system and remain the authoritative scientific documents.

The files in this archive support inspection and offline rerunning of analyses from the included data. Re-acquiring optical measurements requires the original optical platform and instrument-control environment, and is therefore outside the scope of this source-data/code deposit.

Repository Deposit and Journal Upload

Because this source-data package is larger than a typical journal Supplementary Information file, we deposit it on Zenodo and cite the record in the journal submission. The submission route is:

1. Upload `SM_Source_Materials_CORE_5REPORTS.tar.gz` and its checksum file to the Zenodo record associated with the DOI below.
2. Use the same repository DOI in the manuscript, Supplementary Information, this guide, and the Zenodo record.
3. Cite the DOI in the manuscript Data Availability and Code Availability statements.
4. Submit the formal main manuscript and formal Supplementary Information through the journal submission system.
5. If the journal requests a reviewer-facing PDF for the archive, submit this guide PDF as the lightweight source-data/code guide.

Repository DOI: **10.5281/zenodo.19890402**.
Repository URL: https://doi.org/10.5281/zenodo.19890402.

This archive should be cited as the source-data/code and provenance deposit for the study, and should be read alongside the formal Supplementary Information.

Relation to the Main Manuscript and SI

The main manuscript and Supplementary Information carry the final scientific description: the claims, methods narrative, figure assembly, panel labels, and captions. The Zenodo archive has a narrower role. It records the data, scripts, generated outputs, and report-level materials that allow the analyses to be inspected and rerun offline.

The folder `main_text_data_panels/` provides audit replots for selected data-derived main-text panels only: Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i. The PDF/PNG files in that folder are standalone replots made from the included data and scripts. They are not exact visual copies of the final assembled manuscript panels. Differences in crop, color scale, font, layout, panel spacing, normalization display, or selected representative channels/tasks are expected and do not change the role of the underlying data.

The five report PDFs included under `*/report/` are retained as source reports documenting the analysis history and experiment provenance. They do not replace the formal manuscript or SI. If report-level wording, file labels, or figure layouts differ from the final submission documents, the manuscript and SI should be treated as the formal reference, while this package should be used for data/code traceability.

The same boundary is also summarized in `SM_SCOPE_AND_RELATION_TO_MANUSCRIPT_SI.md`.

Package Contents

Path	Contents
`SM_SCOPE_AND_RELATION_TO_MANUSCRIPT_SI.md`	Boundary between this repository archive, the submitted main manuscript, and the submitted SI.
`ZENODO_DEPOSIT_NOTES.md`	Practical notes for repository deposition and DOI linking.
`00_context/`	Qualitative visual documentation of the optical platform.
`01_result2_paper/`	Report 1: initial transmission-matrix reproduction.

`02_result2_refine_paper/`	Report 2: refined self-referenced transmission-matrix experiment.
`03_result3_paper/`	Report 3: majorization-order transport experiment.
`04_result4_paper/`	Report 4: Complex-B bilinear interaction field, XOR showcase, and semantic benchmark.
`05_result4_refine_paper/`	Report 5: refined Complex-B advantage map and sparse-output analyses.
`main_text_data_panels/`	Audit replots for selected main-text data panels Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i; not exact final manuscript panels.
`shared_raw_data/result4_frame_data/`	Full Result4 frame-level CMOS `.npy` arrays shared by Reports 4 and 5.
`provenance/`	File tree, inventory, checksums, section map, figure-source map, and raw-data inclusion map.

Each report section contains four standard subfolders: `report/`, `source_data/`, `scripts/`, and `figures/`.

The video `00_context/visual_documentation/video_qiushi.mp4` is included as qualitative visual documentation of the live Qiushi Engine optical-platform interface, including instrument-status indicators, optical-power monitoring, SLM/CMOS display panels, and a witness-camera view of the physical setup. It is not used as quantitative evidence for any reported result.

The paths listed in this guide are the paths to use for review and offline reruns. Some source reports were generated during the internal analysis workflow; if a report refers to an older path label, the corresponding files should be located through the report-specific folders and provenance files in this archive.

Integrity Check

From the package root, verify the included files:

```
cd SM_Source_Materials_CORE_5REPORTS
sha256sum -c provenance/checksums_sha256.txt --quiet
```

A silent return indicates that the package files match the checksum manifest.

Software Environment

The offline analysis scripts use standard scientific Python packages:

- Python 3.9 or newer
- `numpy`
- `scipy`
- `matplotlib`

- `scikit-learn`

The package does not require the optical hardware-control software for offline analysis. Hardware-control acquisition scripts are intentionally not included.

Recommended shell setup:

```
export MPLBACKEND=Agg
cd SM_Source_Materials_CORE_5REPORTS
```

Quick Reproduction Map

Report section	Data	Main scripts	Generated figures
`01_result2_paper`	`source_data/`	`make_core_figures.py`; `make_core_figures_v2.py`	`figures/`
`02_result2_refine_paper`	`source_data/`	`generate_paper_figures.py`; `fix_tm_filtering.py`	`figures/`
`03_result3_paper`	`source_data/`	`generate_v3_figures.py`; `generate_raw_data_figure.py`; analysis scripts	`figures/`
`04_result4_paper`	`source_data/`; `shared_raw_data/`	XOR, semantic, and compressed-budget scripts	`figures/`
`05_result4_refine_paper`	`source_data/`; `shared_raw_data/`	phase-2, three-bin, advantage-boundary, and sparse-tap scripts	`figures/`

For exact data/code/figure mapping, see `provenance/FIGURE_SOURCE_MAP.tsv`.

For main-text data-panel mapping, see `main_text_data_panels/MAIN_TEXT_DATA_PANEL_SOURCE_MAP.tsv`. This map covers only Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i and includes a visual-status column marking these outputs as audit replots. It does not attempt to reconstruct the full manuscript page layouts, final assembled panels, or schematic panels.

Report-by-Report Reproduction Commands

The commands below are intended for offline reruns. They overwrite or refresh outputs inside the corresponding report section. For a non-destructive review, copy the package to a working directory before running scripts.

Main-Text Data Panels

```
cd SM_Source_Materials_CORE_5REPORTS
python main_text_data_panels/scripts/generate_main_text_data_panels.py
```

This command regenerates standalone audit-replot outputs for Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i under `main_text_data_panels/generated_panels/`. These files support data/code inspection and should not be used as exact visual copies of the final manuscript panels.

Report 1: Initial Transmission-Matrix Reproduction

```
cd SM_Source_Materials_CORE_5REPORTS/01_result2_paper
python scripts/make_core_figures.py
python scripts/make_core_figures_v2.py
```

Inputs are in `source_data/`. Figures are written to `figures/`.

Report 2: Refined Self-Referenced Transmission Matrix

```
cd SM_Source_Materials_CORE_5REPORTS/02_result2_refine_paper
python scripts/generate_paper_figures.py
```

The script uses `source_data/tm_256`, `source_data/focusing`, `source_data/forward_model`, and `source_data/ring_screening`. The optional script `fix_tm_filtering.py` recomputes filtered TM diagnostics and can overwrite filtered arrays.

Report 3: Majorization-Order Transport

```
cd SM_Source_Materials_CORE_5REPORTS/03_result3_paper
python scripts/generate_v3_figures.py
python scripts/generate_raw_data_figure.py
python scripts/analyze_majorization_spectra.py
python scripts/analyze_majorization_spectra_v2.py
```

Inputs are in `source_data/exp_transport_v1`, `source_data/exp_transport_v2`, and `source_data/exp_transport_v3`.

Report 4: Complex-B Bilinear Interaction Field

The full frame-level CMOS arrays are included once under `shared_raw_data/result4_frame_data/`. Commands:

```
cd SM_Source_Materials_CORE_5REPORTS/04_result4_paper
RAW=../shared_raw_data/result4_frame_data

python scripts/analyze_complex_B_xor_showcase.py \
  --data-dir $RAW/exp_bilinear_001/exp_bilinear_001 \
  --output-dir source_data/processed/complex_B_xor_showcase_results

python scripts/analyze_complex_B_semantic.py \
  --data-dir $RAW/exp_complex_B_semantic \
  --output-dir source_data/processed/complex_B_semantic_results

python scripts/generate_semantic_bin10_figures.py
python scripts/generate_semantic_bin10_figures_v2.py
```

```
python scripts/analyze_semantic_compressed_budget.py \
  --data-dir $RAW/exp_complex_B_semantic \
  --output-dir source_data/processed/semantic_compressed_budget_results
```

Report 5: Refined Complex-B Advantage Map

```
cd SM_Source_Materials_CORE_5REPORTS/05_result4_refine_paper
python scripts/analyze_refinement_phase2_fast.py
python scripts/regenerate_xor_complex_scatter_bin10.py
python scripts/analyze_three_bin_sparse.py
python scripts/analyze_advantage_boundary.py
python scripts/analyze_bin_anchored_twin.py
python scripts/analyze_sparse_tap_twin.py
```

These scripts read the shared raw Result4 frame data and write to the local `source_data/processed/` and `figures/` folders.

Expected Reproduction Level

Level	Supported by this package
Report inspection	Yes. Five report PDFs are included in the five report sections.
Main manuscript and SI inspection	Not included in this repository archive. Submit the formal manuscript and SI separately through the journal system.
Figure inspection	Yes. Generated figures are included in each report section.
Main-text data-panel rerun	Yes. Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i are regenerated by `main_text_data_panels/scripts/generate_main_text_data_panels.py`.
Physical-platform visual documentation	Yes. A short MP4 recording is included under `00_context/visual_documentation/`.
Processed-data inspection	Yes. Each report section contains its own `source_data/`.
Script inspection	Yes. Offline Python scripts are included and syntax-checked.
Offline figure/result rerun	Yes, subject to standard Python dependencies and runtime.
Result4 raw-frame reprocessing	Yes. Frame-level CMOS arrays are included in `shared_raw_data/`.
Optical hardware reacquisition	No. This requires the original physical platform and hardware-control stack.

Validation Summary

At package finalization, the repository archive contained:

- Five report-specific sections.
- One main-text data-panel reproduction section for Fig. 2d-g, Fig. 3d-f, and Fig. 4g-i.

- No main-manuscript or Supplementary Information PDFs inside the repository archive.
- One qualitative optical-platform visual-documentation video.
- Full Result4 frame-level raw data under `shared_raw_data/`.
- Twenty-nine offline Python scripts with zero syntax errors.
- A checksum manifest in `provenance/checksums_sha256.txt`.

Contact

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